

Enhancing Edge Relationships to Construct Knowledge Graph for the Entertainment Domain

Internship Work Summary

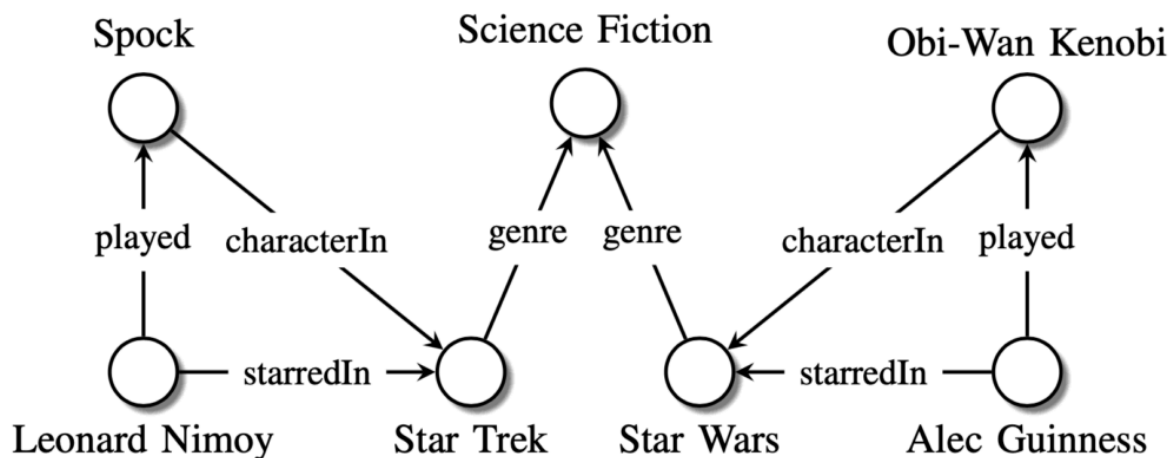
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1. Introduction

Knowledge graphs are advanced data structures that represent information in a network of entities (nodes) and their relationships (edges). They allow for not only the storage of data but also the relationships between them. In a knowledge graph, entities such as people, places, organizations, or concepts are represented as nodes, while the relationships between these entities are depicted as edges [\[1, 2\]](#).

Fig 1. Depiction of a knowledge graph: circles represent nodes, and the arrows between the circles represent the relationships



This project focuses on enhancing the accuracy of predicted relationships between entities and developing a specialized knowledge graph for the entertainment domain. The construction process involves extracting entities and relationships from news summaries, evaluating model predictions, and improving metrics such as Precision, Recall, and F1 score for these relationships. This document provides a comprehensive summary of the project's methodology and the reasoning behind each step.

2. Defining the Problem

2a. Problem Statement

The initial problem statement is to perform two critical tasks effectively:

- 1) Entity Extraction
Entity extraction involves identifying and classifying relevant entities from the text, such as people, movies, organizations, and locations. In the context of the entertainment domain, entities might include actors, directors, production companies, movie titles, and genres[3].
- 2) Relationship Extraction
Once entities are accurately identified, the next step is to extract the relationships between them. Relationship extraction involves determining how entities interact or relate to each other, such as an actor starring in a movie, a director directing a film, or a production company producing a movie [3].

Initially, both tasks have been simultaneously addressed by providing a prompt with few-shot learning to pre-trained LLMs such as Mistral7B, Phi3, and Llama3 for extraction of entities and the relationships between the pairs of entities. On evaluating the extracted entities against the manually annotated ground truth entities on the basis of direct matching, the Mistral7B-v0.2 model gave the best precision, recall and F1-score. We have also further upgraded to Mistral7B-v0.3 and GPT4 LLMs during the later stages of our work, which will be discussed in Section 4. Precision, recall and F1-score had satisfactorily high values for the predicted entities but not for the predicted relationships between pairs of entities, as highlighted in Table 1.

Model		Precision	Recall	F1-Score
Mistral-7B-v0.2	node_name	0.655	0.817	0.683
	node_class	0.687	0.791	0.692
	node_type	0.415	0.529	0.447

	edge_source	0.621	0.655	0.595
	edge_target	0.431	0.436	0.396
	edge_relationship	0.037	0.053	0.041

Table 1. Precision, Recall, F1-Score for Node, Node Class, Node Type, Edge Source, Edge Target, and Edge Relationship on direct matching between ground truth and prediction

2b. Literature survey

Relation extraction (RE) aims to identify semantic relationships between entities in text. Over the years, various techniques have been developed for this purpose [4, 5]. These techniques have been broadly divided into two major categories:

Traditional methods: Rule-based systems leverage predefined rules, while machine learning techniques (unsupervised, semi-supervised, supervised) learn from labelled data. Supervised methods include feature-based and kernel-based approaches.

Deep learning methods: These data-driven techniques learn complex text representations. Supervised learning with architectures like CNNs and RNNs dominates, while distant supervision leverages knowledge bases for training data generation.

We have also explored different text-embedding models present in the literature. Some of the major models are all-MiniLM-L6-v2, all-mpnet-base-v2, etc from the sentence-transformers library of Hugging Face, snowflake-arctic-embed-m-v1.5 from Snowflake and bert-base-uncased from the Flair library. These models have been extensively used for the task of sentence similarity, which makes them potential embedding models for our task. The models are further discussed in Section 3c.

2c. Challenges in our Work

Most of the RE methods discussed in Section 2b fail to perform well in accurate extraction and precise evaluation of relationships from the corpus. The major challenge in our knowledge graph construction that drove this work was the significantly low evaluation metric scores obtained when the predicted relationships by Mistral7B were evaluated against the ground truth relationships. This is because, in many cases, the predicted and the ground truth relationships significantly vary for a given pair of source and target due to ambiguity in the context and interpretation. This results in a clear mismatch while comparing them. This necessitates the development of a novel approach to relation extraction that can address these challenges and lead to a more accurate evaluation process.

3. Methodologies

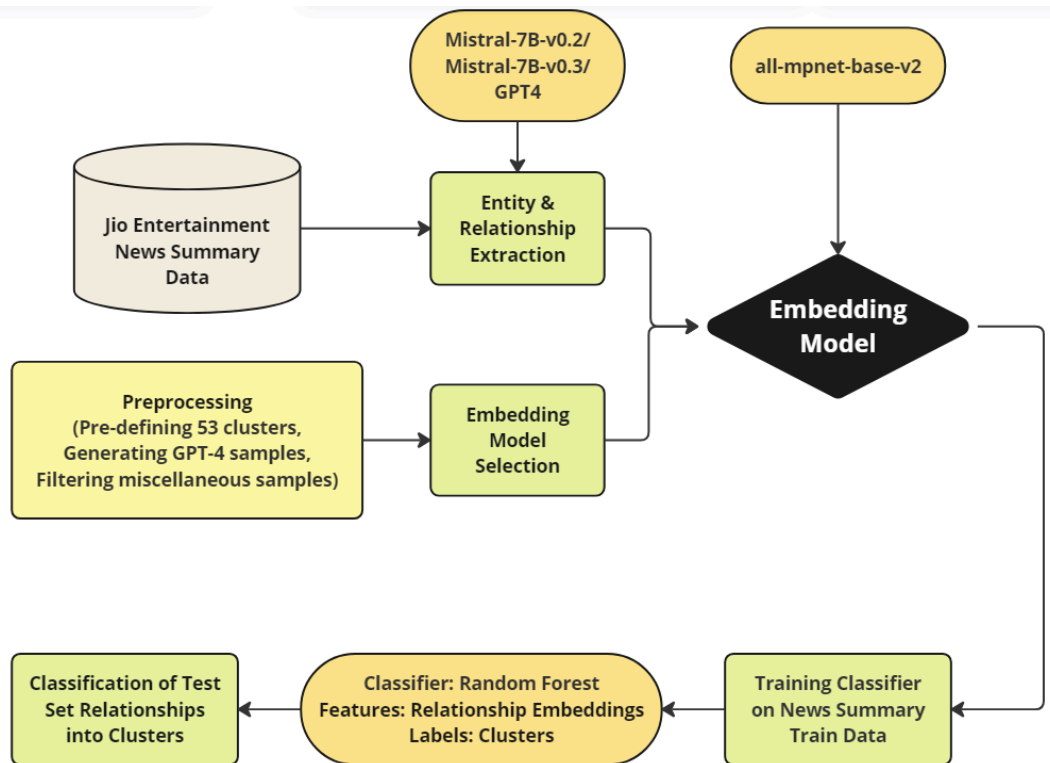


Fig 2. Flowchart depicting the final series of processes for assigning clusters to relationships

3a. Entity and Relationship Extraction

We have used two LLM models (open source - Mistral7B and closed source - GPT4) to generate the entities and relationships. Here, we used prompt-based few-shot learning with three examples of input headline summary and the output as a dictionary with respective keys: nodes and edges. Node has been subdivided into name, class, type and property.

An example of the input and output of the Mistral-7B-v0.2 model is shown:

Input news summary: “Actors Rakul Preet Singh and Jackky Bhagnani will get married on Feb 21 in an intimate ceremony in Goa. They recently kicked off their wedding festivities with a dhol night at Bhagnani's residence. The two initially planned to have their wedding in the Middle East but changed their plans after PM Modi's call to choose India as the venue for big life events.”,

Mistral-7B-v0.2 output:

names (nodes): ['Rakul Preet Singh', 'Jackky Bhagnani', 'Goa', 'Bhagnani's Residence', 'PM Modi'].

node classes: ['Person', 'Person', 'Location', 'Location', 'Person'], node types: ['Actress', 'Actor', 'State', 'Residential Property', 'Politician'],

node properties: [['Getting Married on Feb 21'], ['Getting Married on Feb 21'], [], [], ['Called to choose India as the venue for big life events']],

source nodes: ['Rakul Preet Singh', 'Rakul Preet Singh', 'Rakul Preet Singh', 'Jackky Bhagnani', 'Jackky Bhagnani', 'PM Modi'],

target nodes: ['Jackky Bhagnani', 'Goa', 'Bhagnani's Residence', 'Goa', 'Bhagnani's Residence', 'Rakul Preet Singh & Jackky Bhagnani'],

edge relationships: ['Fiance', 'Getting Married In', 'Had Dhol Night At', 'Getting Married In', 'Had Dhol Night At', 'Asked to Choose India As Venue'].

3b. Pre-processing

In order to evaluate the relationship field for the entertainment domain data, we performed a series of pre-processing steps which are explained further in detail.

Step 1: As a starting point, 53 clusters were manually pre-defined, along with cluster names and numbering, as shown in the table below.

1	2	3	4	5	6	7
DEATH & MOURNING	Legal and Criminal Proceedings	Musical Relationships and Associations	Connectivity	Appearances and Interaction	Interpersonal Relations	Teaser Preview and sharing
Deceased	Charges fine	Band Member	Affiliation	Meeting with	Mother	Shares trailer
Mourns	Arrested with	Former bandmate	Connected with	Conversation with	Mother-daughter	Leaked Clips
Murder Victim	Banned	Tiff with former bandma	Linked to	Interviewer	Mother-Son	Shared video of
Final rites	Contests bail	Playmate	Relative	Interviewed	Son of	Showcase
Location of death	Seized during raid	Fan club	Connected to	Interviewed by	Daughter	Previewed
Found dead in	Lifted ban	Bandmate	Connecting factc	Guest on KBC with*	Daughter of	Shares sneak pee
Funeral in	Dismissed plea	Band member		Interacted with*	Father of	video of
Laid to rest at	Adjourned			Interaction*	Mother of	Teaser attached v
Died at	Detained in			Appeared on KBC with*	Daughter-in-law	Teaser Unveiling
Memorial	Cancelled Arrest Warrant			Special guest	Father	
Death Place	Met in prison			Extended cameo	Father-in-law	
Deceased person*	Quashed case			Guest on show	Stepmother	

Table 2: First 7 Cluster Samples from the Pre-defined 53 Clusters

Step 2: The LLM model was used to generate new relationships by giving appropriate prompts in order to further expand the samples in each cluster and increase cluster diversity. These test samples were used to select the best embedding model, which is discussed in Section 3b.

Step 3: We performed another clustering iteration manually to clean the out-of-cluster/ miscellaneous samples in each cluster to either discard them or add them to the closest clusters based on their importance with respect to the entertainment domain.

3c. Embedding Model Selection

We majorly worked with 5 models for creating embeddings of the relationships:

1. **all-MiniLM-L6-v2:** A sentence-transformers model with 22.7M parameters based on the MiniLM architecture, a smaller and faster alternative to BERT. It maps sentences &

paragraphs to a 384-dimensional dense vector space. The model is intended to be used as a sentence and short paragraph encoder. Given an input text, it outputs a vector that captures the semantic information. The sentence vector may be used for information retrieval, clustering or sentence similarity tasks.

2. **paraphrase-MiniLM-L6-v2**: A sentence-transformers model with 22.7M parameters that is built upon all-MiniLM-L6-v2 but is specifically trained for paraphrase identification. It maps sentences & paragraphs to a 384-dimensional dense vector space.
3. **all-mpnet-base-v2**: A sentence-transformers model with 109M parameters based on the Multi-scale Perceptron Transformer (MBERT) architecture. It maps sentences & paragraphs to a 768-dimensional dense vector space.
4. **Snowflake-arctic-embed-m**: A model by Snowflake with 110M parameters that maps sentences & paragraphs to a 768-dimensional dense vector space.
5. **bert-base-uncased (Flair)**: A pre-trained BERT model (Bidirectional Encoder Representations from Transformers) using the Flair library. It has 110M parameters and creates a 768-dimensional dense vector space. It is a powerful model for understanding contextual relationships in text.

3d. Proposed Methods/Algorithms

To make the ground truth and Mistral7B predicted relationships match better and improve the evaluation metric scores for the relationships, we have followed 3 major approaches that take the semantic meaning of the relationships into account:

a. Clustering

Instead of evaluating the Mistral7B predicted relationships directly on a one-on-one basis with the ground truth relationships, evaluating the predictions based on whether they fall in the same cluster as the ground truth is considered (Cluster-Based Metrics). This method considers the semantic similarity and contextual alignment between relationships, providing a more robust and meaningful evaluation metric and reflects the true performance of the model.

To evaluate the performance metrics on the relationship field, we performed a series of iterative steps, as detailed below:

Iteration 1: Assigning Clusters to Set 1 Relationships

In this phase, we focused on assigning clusters to a set of relationships extracted from 81 news summaries. These relationships included manually annotated ground truth and Mistral7B

predicted entities, along with their classes, types, properties, and the relationships between them. Each relationship had a corresponding source and target entity.

- **Set 1 Dataset Description:**

1. **News Summaries:** 81 summaries.
2. **Entities:** Manually annotated ground truth and Mistral7B predicted entities.
3. **Entity Details:** Classes, types, and properties of entities.
4. **Relationships:** Edges between entities, detailing the type of relationship and its context.

- **Cluster Assignment Process:**

1. **Embedding Calculation:**

- The paraphrase-MiniLM-L6-v2 model, chosen for its superior performance in previous trials, was used to generate embeddings for each pair of ground truth and predicted relationships.

2. **Mean Embedding Calculation:**

- For each of the 53 pre-defined clusters, mean embeddings were calculated for each cluster based on the embeddings of the relationships already assigned to them.

3. **Cosine Similarity-based Cluster Assignment:**

- Cosine similarity was computed between the embedding of each relationship and the mean embeddings of the clusters.
- Both ground truth and predicted relationships were assigned to the cluster with the highest cosine similarity.
- This assignment provided a foundation for further clustering iterations and refinements.

Iteration 2: Assigning Clusters to GPT-4 Relationships and Evaluation

In this phase, we assigned clusters to the relationships generated by GPT-4 and evaluated the performance metrics of these assignments using various embedding models (Results in 4b.).

- **Assignment of Ground Truth Clusters:**

- Initially, the GPT-4 generated relationships were assigned to specific clusters, which were recorded as the ground truth clusters for each generated test sample.
- These relationships were then extracted from their initial clusters to be reassigned based on their embeddings.

- **Cosine Similarity-based Cluster Reassignment:**

- The reassignment process involved calculating the cosine similarity between the embeddings of the generated relationships and the mean embeddings of the clusters.
- The GPT-4 generated relationships were reassigned to the Top-5 clusters that had the highest cosine similarity scores.
- These updated clusters with reassigned relationships are used for further clustering iterations and refinements.
- **Evaluation of Cluster Assignments:**
 - The evaluation involved comparing the ground truth clusters with the clusters predicted by the embedding models.
 - **Ground Truth:** Only the top 1 predicted cluster was considered for ground truth relationships.
 - **Predictions:** The top 5 predicted clusters were considered for the predicted relationships.
 - **Metrics:** Precision, recall, F1-score, and accuracy were calculated for the top 1, top 3, and top 5 predicted clusters for each predicted relationship.
 - We considered various embedding models for this task, as discussed in Section 3c, and performed evaluations by considering the top 1, top 3, and top 5 predicted clusters for each GPT-4 generated relationship.
 - If the ground truth cluster is present in the Top 1 predicted cluster, it would be considered True Positive for the Top 1 case. If the ground truth cluster is present in the Top 3 predicted clusters, it would be considered True Positive (TP) for the Top 3 cases, and the same goes for the Top 5 cases. The conditions for calculating TP, FP, FN, and TN, along with the results, are discussed in Section 4.

Iteration 3: Assigning Clusters to Set 2 Relationships and Evaluation

In this phase, we focused on a new set of 100 news summaries, evaluating the effectiveness of cluster assignment using different statistical methods for embeddings (Results in 4c. prior thresholding).

- **Set 2 Dataset Description:**
 1. **News Summaries:** 100 summaries.
 2. **Entities:** Manually annotated ground truth and Mistral7B predicted entities.
 3. **Details:** Classes, types, properties, and relationships with corresponding source and target entities.
- **Data Processing and Considerations:**
 - Only relationships with common pairs of source and target entities in both ground truth and predictions were considered for clustering.

- Relationships without common pairs were excluded from this evaluation. For a fair evaluation, it's crucial to compare predictions with the corresponding ground truth. Relationships with common source-target pairs ensure that we are comparing similar or identical entities and relationships, providing a consistent basis for evaluation.

- **Cluster Assignment Process:**

1. **Embedding Model:**

- From the evaluation results of Iteration 2, the all-mpnet-base-v2 model gave the best Precision/Recall/F1-score values.
- Hence, we continue to use the all-mpnet-base-v2 model for creating embeddings in this iteration and in the rest of our work.

2. **Embedding Calculation:**

- The all-mpnet-base-v2 model was used to generate embeddings for each pair of ground truth and predicted relationships.

3. **Mean and Median Embedding Calculation:**

- For each of the 53 pre-defined clusters, mean and median embeddings were calculated based on the embeddings of the relationships already assigned to them.
- We experimented with different statistical methods, i.e., mean and median embeddings of the clusters, to evaluate the accuracy of the embeddings of the nearest clusters for a given test relationship.
- Clustering was performed using both mean and median embeddings to evaluate which statistical method provided better clustering results.

4. **Cosine Similarity:**

- Cosine similarity was computed between the embedding of each relationship and both the mean and median embeddings of the clusters.
- Each predicted relationship is assigned Top 5 clusters based on cosine similarity scores.

5. **Cluster Assignment:**

- If the cosine similarity between the ground truth and the predicted relationship exceeds 0.75 (similarly, tried with 0.4 as well; results in 4e. and 4f. respectively), the predicted relationship was directly assigned to the ground truth cluster. In this case, the predicted Top 1 cluster would be the same as the ground truth cluster and the rest of the Top 4 clusters would be clusters having the next highest cosine similarity scores.
- If the above condition is not satisfied, the predicted relationship would be assigned all the Top 5 clusters based on the highest cosine similarity scores in the decreasing order.

- **Evaluation of Cluster Assignments:**

- **Ground Truth:** Only the top 1 predicted cluster was considered for ground truth relationships.
- **Predictions:** The top 5 predicted clusters were considered for the predicted relationships.
- **Metrics:** Precision, recall, F1-score, and accuracy were calculated for the top 1, top 3, and top 5 predicted clusters for each predicted relationship.
- Evaluation was done for both mean and median embeddings-based clustering.
- The results are discussed in Section 4.

b. Classification

Iteration 4: Assigning Clusters to Set 2 Relationships by Classification

Another approach was explored, in which we treated the cluster assignment as a supervised classification task. Different classifiers, namely, KNN, Random Forest, SVM, and XGBoost, have been trained with the previous relationships embeddings as the training features and the corresponding clusters as the training labels (Results in 4d.).

- **Training Data:** The training was conducted on the Set 1 relationships (comprised of 81 news summaries with manually annotated ground truth and Mistral7B predicted entities and relationships) and the GPT-4 generated relationships, as we already had the cluster labels for these relationships from previous iterations. The embeddings of these relationships, along with their cluster labels, were used to train the classifiers.
- **Test Data:** The embeddings of the ground truth and predicted relationships from the new set of 100 summaries (Set 2) were used as the test sample features, and their clusters were predicted using the trained classifier models.
- **Classification Process:**
 - If the cosine similarity between the ground truth and the predicted relationship exceeds 0.75 (similarly, tried with 0.4 as well; results in 4e. and 4f. respectively), the predicted relationship was directly assigned to the ground truth cluster. In this case, the predicted Top 1 cluster would be the same as the ground truth cluster and the rest of the Top 4 clusters would be clusters having the next highest cosine similarity scores.
 - If the above condition is not satisfied, the predicted relationship would be assigned all the Top 5 clusters based on the classification output.
 - The classifier models are trained on the training data, where they learn the embeddings of relationships as features and corresponding clusters as the class labels.

- For the test data, given the embeddings of test relationships as the input features, the classifier model would give the probabilities of the given relationship belonging to each cluster.
 - For the ground truth relationships, the cluster with the highest probability is taken.
 - For the predicted relationships, instead of classifying the test relationship as belonging to the cluster with the highest probability, we take the 5 clusters with the Top 5 probabilities as the prediction outcomes of the classifier models.
- **Evaluation of Cluster Assignments:**
 - **Ground Truth:** Only the top 1 predicted cluster was considered for ground truth relationships.
 - **Predictions:** The top 5 predicted clusters were considered for the predicted relationships.
 - **Metrics:** Precision, recall, F1-score, and accuracy were calculated for the top 1, top 3, and top 5 predicted clusters for each predicted relationship.

c. Cosine Similarity-based Cluster Assignment

As a third approach, we also tried to assign clusters to relationships by defining a threshold for the cosine similarity between a given relationship and the mean cluster embeddings. The idea was to assign the relationship to the cluster that has cosine similarity above the threshold. We tried to define Cluster 1 Threshold, Cluster 2 Threshold, and so on until Cluster 5 Threshold, such that if the cosine similarity between a given relationship and the mean cluster embedding exceeds the Cluster 1 Threshold, it would be assigned that cluster as the Top 1 cluster. Top 2, Top 3, Top 4, and Top 5 clusters would also be assigned in a similar way for a given relationship.

Threshold-Based Cluster Assignment:

- **Objective:** Assign a relationship to the cluster whose mean embedding has a cosine similarity above a defined threshold.
- **Thresholds:** Define Cluster 1 Threshold, Cluster 2 Threshold, up to Cluster 5 Threshold.
 - If the cosine similarity between a given relationship and the mean cluster embedding exceeds the Cluster 1 Threshold, it is assigned that cluster as the Top 1 cluster.
 - The same logic applies to assigning Top 2, Top 3, Top 4, and Top 5 clusters.

Hypothesis:

- Cosine similarities with the Top 1 clusters and Top 2 clusters were expected to have a clear distinction in their ranges.
- The Cluster 1 Threshold was to be defined such that: Cosine similarities for all Top 1 clusters > Cluster 1 Threshold > Cosine similarities for all Top 2 clusters.

- The same logic was expected to apply for Top 2, Top 3, Top 4, and Top 5 clusters.

Outcome:

- **Failure of Threshold-Based Approach:**
 - In practice, all the Top 1 cluster similarities, Top 2 cluster similarities, and so on had overlapping ranges of cosine similarity scores.
 - There was no clear distinction in the similarity scores for us to define thresholds between them.
 - Consequently, it was not possible to set a Cluster 1 Threshold that separated Top 1 cluster similarities from Top 2 cluster similarities, and so forth.

Conclusion:

The threshold-based cluster assignment approach was not viable in our case due to the overlapping ranges of cosine similarity scores for the Top 1, Top 2, Top 3, Top 4, and Top 5 clusters. Our assumption of distinct ranges for different top clusters did not hold true, hence rendering this approach ineffective for our dataset.

3e. Best Proposed Algorithm

In our exploration of relation extraction on the private news summaries dataset, we experimented with various models and techniques for both embedding generation and classification. After thorough analysis and multiple iterations, we identified the optimal combination of methods that yielded the best results. This section outlines the best-performing approaches and models used in our approach.

Embedding Generation

For the task of embedding generation, the **MPNet model** was chosen after comparing several pre-trained models, including BERT, MiniLM, and Snowflake's Arctic. The comparison of evaluation metrics between different embedding models is shown in section 4b. MPNet stood out by consistently producing higher-quality embeddings to capture contextual relationships within sentences, which is crucial for relation extraction tasks.

Clustering Approach

For the task of assigning clusters to relationships, we mainly evaluated three primary approaches: mean-based clustering, median-based clustering, and a direct classification approach using machine learning models. Among these, the direct classification approach proved superior.

1. Clustering Approaches:

- **Mean-Based Clustering:** This method involves calculating the mean embeddings for each class/cluster and assigning new samples to the cluster with the closest

mean. While simple, it struggled with the nuanced distinctions required for accurate relation extraction.

- **Median-Based Clustering:** Similar to the mean-based approach, this method calculates the median embedding for each class. However, it too proved to be less effective. The comparison of results between mean and median clustering approaches is shown in section 4c.

2. Classification Approach:

- **Random Forest Classifier:** The classification approach that outperformed all others was based on training a **Random Forest (RF) classifier** on the embeddings of the relationships of the Set 1 dataset, generated from the MPNet model.
- **Performance of Random Forest:** The RF classifier demonstrated superior performance compared to other classifiers such as Support Vector Machines (SVM), Logistic Regression, and K-nearest neighbours (KNN). The comparison of results between classification models is shown in section 4d.

3. Results:

- The classification approach using RF outperformed the clustering-based approaches by a significant margin. The RF model achieved the highest F1 scores.

Application on Private and Public Datasets

- **Jio News Summary Dataset:** Initially, trained, validated, and employed our combination of best models/algorithms using a private dataset containing news summaries from Jio. This dataset, being complex and unstructured, allowed us to fine-tune our methods and come up with ways to handle the complexity.
- **CoNLL04 Public Dataset:** To validate the generalizability of our approach, we applied the same algorithm to the CoNLL04 dataset, a standard benchmark in relation extraction tasks. Despite the differences in data structure and content between the private and public datasets, our approach, using the RF classifier with MPNet embeddings, demonstrated a strong performance.

Entity and Relation Extraction

For the task of entity and relation extraction from news summaries, we experimented with Mistral-7B-v0.2, Mistral-7B-v0.3 and GPT4. Each model outperformed the previous model in relation extraction. Sections 4g and 4h show the better performance of Mistral-7B-v0.3 over Mistral-7B-v0.2 when RF classification approach is employed with different cosine similarity thresholds. Section 4i shows that GPT4 further outperformed Mistral-7B-v0.3 when RF classification approach is applied on the CoNLL04 dataset (discussed in Section 4).

4. Applying Classification Approach on CoNLL04 Dataset

Now that the best algorithm has been found to be the classification approach, we tried the same on the CoNLL04 public dataset [\[7\]](#). The dataset contains sentences annotated with entities (e.g., people, organizations, locations) and relationships between them. The goal is to evaluate a model's ability to predict these relationships using embeddings and a Random Forest (RF) classifier, which has given the best results so far. The results obtained in this step are compared with those of the private dataset and also with results from other similar works that worked on that same CoNLL04 dataset (Section 4i and 4j)

Dataset Preparation:

The CoNLL04 dataset includes three subsets: 922 training samples, 231 validation samples, and 288 test samples. Each subset contains:

- **Sentences:** Textual data with tokens representing entities and relationships.
- **Entities:** Pairs of subjects and objects within the sentence.
- **Relationships:** Annotations indicating the type of relationship between the entities, such as Located_In, Work_For, OrgBased_In, Live_In, and Kill.

Predicted Relations Using Mistral7B and GPT4:

- Used the Mistral7B-v0.3 and GPT4 model to extract relationships by feeding sentences and the corresponding source-target pairs into the prompt for the train, validation and test sets.
- The outputs were saved in separate CSV files containing predicted relations for three files: train, validation, and test.
- Merged the ground truth dataset and the predicted relations dataset based on common subject-object pairs to create a new dataset that includes columns for:
 - sentence
 - subject
 - object
 - ground truth relation
 - predicted relation

Feature Extraction:

1. **Label Encoding:**
 - Encoded the ground truth relations as integer labels: Located_In, Work_For, OrgBased_In, Live_In, and Kill mapped to labels 0, 1, 2, 3, and 4, respectively.
2. **Embedding Generation:**

- Used the all-mpnet-base-v2 model to generate embeddings for the predicted relations. This was done by feeding the predicted relations into the model and extracting the corresponding embeddings.

Training the Classifier:

1. Model Selection:

- Chose a Random Forest (RF) classifier for the task due to its robustness in the previous experiments and ability to handle high-dimensional data (like embeddings).

2. Training:

- Trained the RF classifier on the embeddings generated from the predicted relations. The class labels were the corresponding ground truth relations from the training data.

3. Validation:

- Evaluated the model using the validation dataset, which contains labelled samples not used in training. This provided an estimate of the model's performance on unseen data.

Evaluation:

1. Testing:

- Also applied the trained RF classifier to the test data. For each test sample, the embedding of the predicted relation was fed into the classifier to predict the relation label.

2. Performance Metrics:

- **Micro F1 Score:** Calculated the micro F1 score, which aggregates the contributions of all classes to compute an overall F1 score. Provides a measure of overall performance across all relation types, accounting for any class imbalance in the dataset.
- **Macro F1 Score:** Calculated the macro F1 score, which computes the F1 score for each class individually and then averages them. This metric gives equal weight to each class, regardless of its frequency. Illustrates the model's performance across each class independently, highlighting how well the model performed on both frequent and rare classes.

5. Experiments and Results

4a. Performance metrics evaluated

True Positive (TP): Number of correctly predicted positive samples for a specific class, i.e., when the ground truth cluster is within the top 5 predicted clusters.

True Negative (TN): Number of correctly predicted negative samples for a specific class. For each ground truth and predicted cluster pair, we counted true negatives for each cluster number that is not the ground truth and is also not in the predicted clusters.

False Positive (FP): Number of samples incorrectly predicted as positive for a specific class. $FP = 0$ if the ground truth cluster is within the top 5 predicted clusters and there are 4 wrongly predicted clusters. $FP = 1$ if the ground truth cluster is not within the top 5 predicted clusters and there are 5 wrongly predicted clusters.

False Negative (FN): Number of samples incorrectly predicted as negative for a specific class, i.e., when the ground truth cluster is not within the top 5 predicted clusters.

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Accuracy} = \frac{\text{True Positives (TP)} + \text{True Negatives (TN)}}{TP + TN + FP + FN}$$

4b. Iteration 2 Mean Clustering Approach Results

Different embedding models were compared based on their performance in the clustering task of GPT-4 extracted relationships. The all-mpnet-base-v2 model consistently outperformed other models, giving the best F1 scores for all top k comparisons.

Model	Top 5		Top 3		Top 1	
all-mpnet-base-v2	TP	2544	TP	2410	TP	1876
	FP	219	FP	353	FP	887
	F1-score	0.921	F1-score	0.872	F1-score	0.679

snowflake-arctic -embed-m	TP	2146	TP	1957	TP	1458
	FP	617	FP	806	FP	1305
	F1-score	0.777	F1-score	0.708	F1-score	0.527
all-MiniLM-L6-v2	TP	2458	TP	2295	TP	1717
	FP	305	FP	468	FP	1046
	F1-score	0.889	F1-score	0.831	F1-score	0.621
paraphrase-MiniLM -L6-v2	TP	2432	TP	2253	TP	1650
	FP	331	FP	510	FP	1113
	F1-score	0.88	F1-score	0.815	F1-score	0.597
bert-flair	TP	1372	TP	1120	TP	690
	FP	1391	FP	1643	FP	2073
	F1-score	0.496	F1-score	0.405	F1-score	0.249

4c. Iteration 3 Mean/Median Clustering Approach Results

Mean and median clustering approaches were applied to the embeddings of the Set2 relationships (100 news summaries), with all-mpnet-base-v2 showing superior results in terms of accuracy and F1-score.

Model	Top 5		Top 3		Top 1	
all-mpnet-base-v2 (mean embeddings)	TP	71	TP	61	TP	43
	FN	24	FN	34	FN	52
	FP	404	FP	224	FP	52
	TN	4536	TN	4716	TN	4888
	Precision	0.149	Precision	0.214	Precision	0.452
	Recall	0.747	Recall	0.642	Recall	0.452
	F1 score	0.249	F1 score	0.321	F1 score	0.452
	Accuracy	0.915	Accuracy	0.949	Accuracy	0.979

all-mpnet-base-v2 (median embeddings)	TP	64	TP	56	TP	42
	FN	31	FN	39	FN	53
	FP	411	FP	229	FP	53
	TN	4529	TN	4711	TN	4887
	Precision	0.135	Precision	0.196	Precision	0.442
	Recall	0.673	Recall	0.589	Recall	0.442
	F1 score	0.224	F1 score	0.295	F1 score	0.442
	Accuracy	0.912	Accuracy	0.947	Accuracy	0.979

4d. Iteration 4 Classification Approach Results

Various classification algorithms were tested, including KNN, RF, SVM, and XGB, with RF consistently delivering the best performance in terms of precision, recall, and accuracy.

Model	Top 5		Top 3		Top 1	
all-mpnet-base-v2 (knn classification)	TP	64	TP	58	TP	48
	FN	31	FN	37	FN	47
	FP	411	FP	227	FP	47
	TN	4616	TN	4733	TN	4893
	Precision	0.135	Precision	0.203	Precision	0.505
	Recall	0.674	Recall	0.611	Recall	0.505
	F1 score	0.224	F1 score	0.305	F1 score	0.505
	Accuracy	0.914	Accuracy	0.948	Accuracy	0.981
all-mpnet-base-v2 (RF classification)	TP	70	TP	65	TP	50
	FN	25	FN	30	FN	45
	FP	405	FP	220	FP	45
	TN	4535	TN	4720	TN	4895
	Precision	0.147	Precision	0.228	Precision	0.526

	Recall	0.737	Recall	0.684	Recall	0.526
	F1 score	0.245	F1 score	0.342	F1 score	0.526
	Accuracy	0.914	Accuracy	0.95	Accuracy	0.982
all-mpnet-base-v2 (SVM classification)	TP	74	TP	62	TP	48
	FN	21	FN	33	FN	47
	FP	401	FP	223	FP	47
	TN	4539	TN	4717	TN	4893
	Precision	0.156	Precision	0.217	Precision	0.505
	Recall	0.779	Recall	0.653	Recall	0.505
	F1 score	0.259	F1 score	0.326	F1 score	0.505
	Accuracy	0.916	Accuracy	0.949	Accuracy	0.981
all-mpnet-base-v2 (XGB classification)	TP	66	TP	60	TP	44
	FN	29	FN	35	FN	51
	FP	409	FP	225	FP	51
	TN	4531	TN	4715	TN	4889
	Precision	0.139	Precision	0.21	Precision	0.463
	Recall	0.695	Recall	0.632	Recall	0.463
	F1 score	0.231	F1 score	0.316	F1 score	0.463
	Accuracy	0.913	Accuracy	0.948	Accuracy	0.978

4e. Post 0.75 Thresholding Results for Mistral7B-v0.2 Predicted Relationships

Model	Top 5		Top 3		Top 1	
	TP	71	TP	63	TP	48
	FN	24	FN	32	FN	46

all-mpnet-base-v2 (mean clustering)	FP	24	FP	32	FP	46
	TN	4536	TN	4718	TN	4894
	Precision	0.747	Precision	0.663	Precision	0.516
	Recall	0.747	Recall	0.663	Recall	0.516
	F1 score	0.747	F1 score	0.663	F1 score	0.516
	Accuracy	0.989	Accuracy	0.987	Accuracy	0.982
all-mpnet-base-v2 (RF classification)	TP	71	TP	67	TP	57
	FN	24	FN	28	FN	38
	FP	24	FP	28	FP	38
	TN	4536	TN	4722	TN	4902
	Precision	0.747	Precision	0.705	Precision	0.6
	Recall	0.747	Recall	0.705	Recall	0.6
	F1 score	0.747	F1 score	0.705	F1 score	0.6
	Accuracy	0.989	Accuracy	0.988	Accuracy	0.985
all-mpnet-base-v2 (SVM classification)	TP	74	TP	63	TP	50
	FN	21	FN	32	FN	45
	FP	21	FP	32	FP	45
	TN	4539	TN	4718	TN	4895
	Precision	0.779	Precision	0.663	Precision	0.526
	Recall	0.779	Recall	0.663	Recall	0.526
	F1 score	0.779	F1 score	0.663	F1 score	0.526
	Accuracy	0.991	Accuracy	0.987	Accuracy	0.982

4f. Post 0.4 Thresholding Results for Mistral7B-v0.2 Predicted Relationships

Model	Top 5		Top 3		Top 1	
all-mpnet-base-v2 (mean clustering)	TP	74	TP	69	TP	63
	FN	21	FN	26	FN	32
	FP	21	FP	26	FP	32
	TN	4539	TN	4724	TN	4908
	Precision	0.779	Precision	0.726	Precision	0.663
	Recall	0.779	Recall	0.726	Recall	0.663
	F1 score	0.779	F1 score	0.726	F1 score	0.663
	Accuracy	0.991	Accuracy	0.989	Accuracy	0.987
all-mpnet-base-v2 (RF classification)	TP	76	TP	66	TP	66
	FN	19	FN	29	FN	29
	FP	19	FP	29	FP	29
	TN	4541	TN	4911	TN	4911
	Precision	0.8	Precision	0.695	Precision	0.695
	Recall	0.8	Recall	0.695	Recall	0.695
	F1 score	0.8	F1 score	0.695	F1 score	0.695
	Accuracy	0.992	Accuracy	0.988	Accuracy	0.988
all-mpnet-base-v2 (SVM classification)	TP	76	TP	69	TP	64
	FN	19	FN	26	FN	31
	FP	19	FP	26	FP	31
	TN	4541	TN	4724	TN	4909
	Precision	0.8	Precision	0.726	Precision	0.674
	Recall	0.8	Recall	0.726	Recall	0.674
	F1 score	0.8	F1 score	0.726	F1 score	0.674

	Accuracy	0.992	Accuracy	0.989	Accuracy	0.987
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4g. Post 0.75 Thresholding Results for Mistral7B-v0.3 Predicted Relationships

We also tried the **Mistral7B-v0.3** model for the extraction of entities and relationships and applied the classification approach, as it showed the best results in the previous iterations. As a result, an improvement in the performance was seen over the **Mistral7B-v0.2** model. The table below shows results when a cosine similarity threshold of 0.75 was applied.

Model	Top 5		Top 3		Top 1	
Mistral7B-v0.3 all-mpnet-base-v2 (RF classification)	TP	105	TP	100	TP	89
	FN	37	FN	42	FN	53
	FP	37	FP	42	FP	53
	TN	6779	TN	7058	TN	7331
	Precision	0.739	Precision	0.704	Precision	0.627
	Recall	0.739	Recall	0.704	Recall	0.627
	F1 score	0.739	F1 score	0.704	F1 score	0.627
	Accuracy	0.989	Accuracy	0.988	Accuracy	0.986

4h. Post 0.4 Thresholding Results for Mistral7B-v0.3 Predicted Relationships

The table below shows the results when a cosine similarity threshold of 0.4 was applied to the **Mistral7B-v0.3** extracted relationships. From the results, we can conclude that a threshold of 0.4 is an assumption after manual checking and satisfactory results.

Model	Top 5		Top 3		Top 1	
Mistral7B-v0.3 all-mpnet-base-v2 (RF classification)	TP	114	TP	112	TP	107
	FN	28	FN	30	FN	35
	FP	28	FP	30	FP	35
	TN	6788	TN	7070	TN	7349
	Precision	0.803	Precision	0.788	Precision	0.753

	Recall	0.803	Recall	0.788	Recall	0.753
	F1 score	0.803	F1 score	0.788	F1 score	0.753
	Accuracy	0.992	Accuracy	0.992	Accuracy	0.991

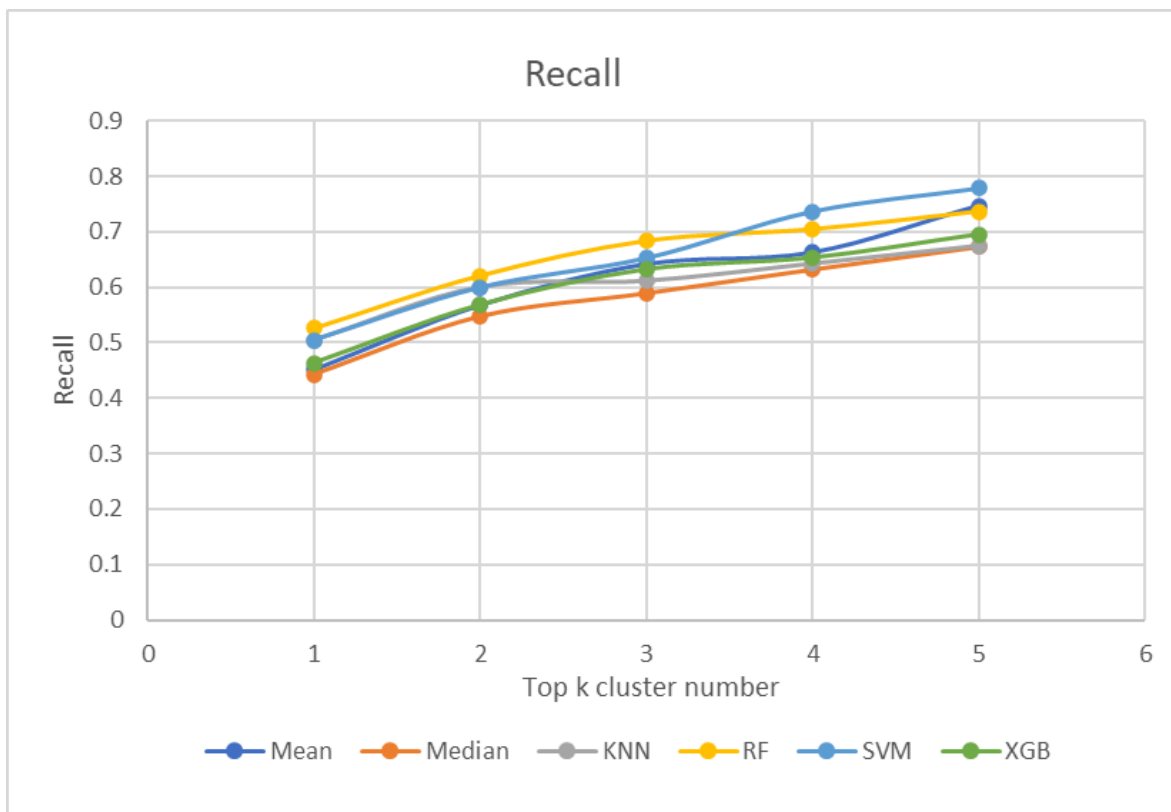
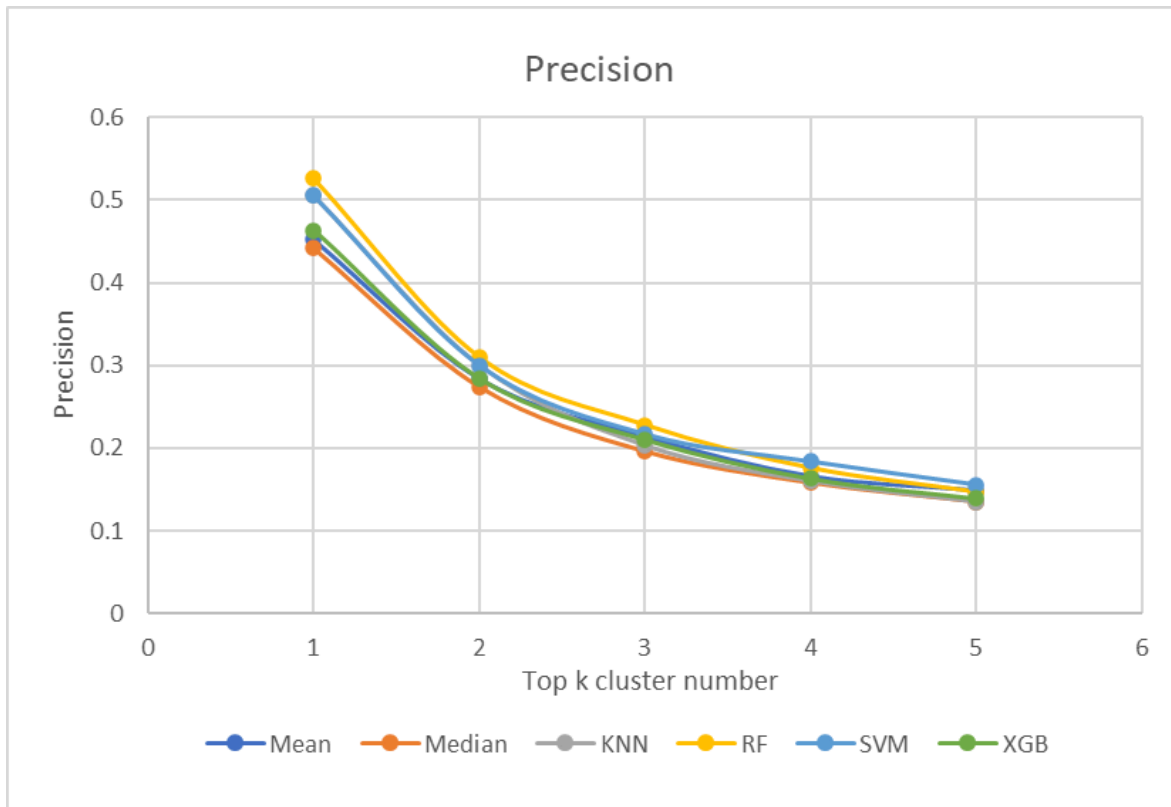
4i. Comparison of Private and CoNLL04 Dataset Results

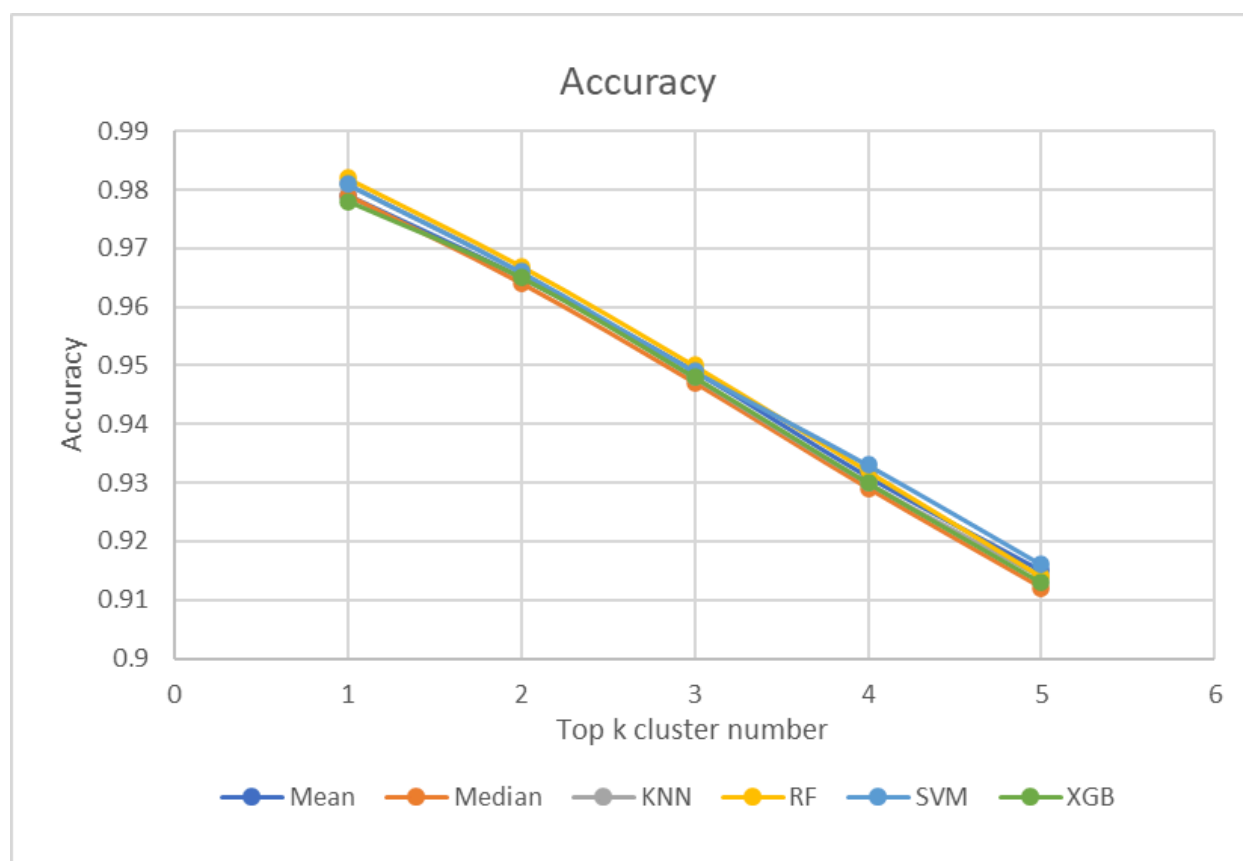
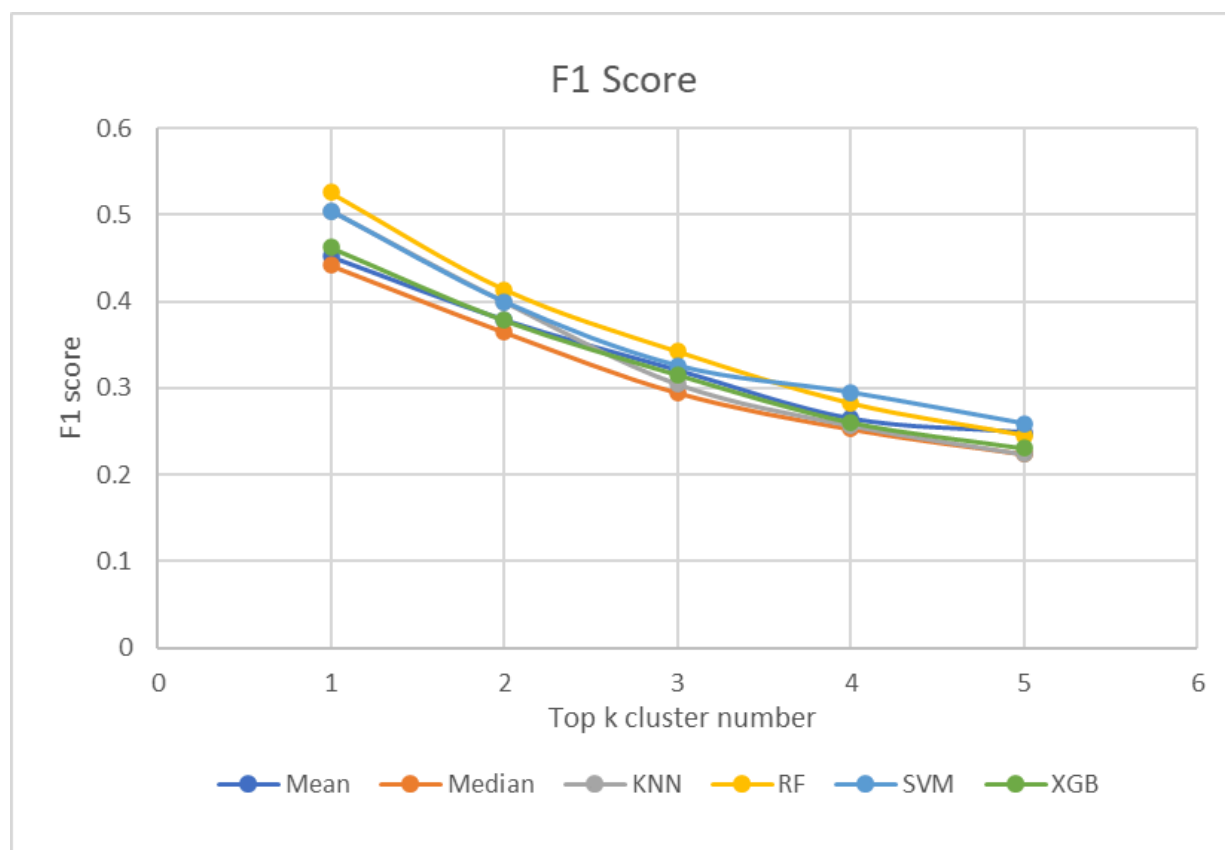
Dataset		Mistral-7B-v0.3		GPT4	
		F1-score (Micro)	F1-score (Macro)	F1-score (Micro)	F1-score (Macro)
Jio News Summaries (Private Dataset)	Set 2 Dataset	0.803		0.803	
CoNLL04 (Public)	Validation Dataset	0.778	0.788	0.788	0.808
	Test Dataset	0.715	0.716	0.799	0.799

4j. Comparison of our CoNLL04 Results with Other Works

Works on CoNLL04	F1-score (Micro)	F1-score (Macro)
GPT4+MPNet+RF Classification (Our Approach)	79.9	79.9
REBEL: Relation Extraction By End-to-end Language Generation [8]	75.4	76.65
Autoregressive Structured Prediction with Language Models [9]	76.3	-

4k. Precision, Recall, F1-Scores, Accuracy as a Function of Top K Cluster Number





6. Novelty of Our Approach

Our approach to relation extraction is novel in several key aspects that differentiate it from traditional methods and previous research.

- We leverage **MPNet embeddings**, which offer a significant improvement over traditional embeddings like BERT in better capturing the contextual dependencies between words.
- While most traditional approaches rely on neural networks or clustering methods, by using an RF **classification model** over black-box neural networks, we are able to provide more interpretable results.
- We integrate predicted relations from generative models like **Mistral7B** and **GPT4** into our workflow. By using the LLMs' context-based relationship predictions and comparing them with ground truth relations, we refine our classifier to better capture nuances in the data. This approach merges the predictive strengths of large language models with the accuracy of supervised classifiers, creating a novel pipeline for relation extraction.
- Our methodology also includes an extensive evaluation framework, where multiple algorithms and strategies are compared against each other in a systematic manner. This involves using micro and macro F1 scores to assess performance across different relations, ensuring that our approach is not only effective but also generalizable across different types of relationships.

7. Conclusion

In this study, we developed a comprehensive approach to relation extraction using both private and public datasets. Our methodology uniquely integrates embeddings from the MPNet model and predicted relations generated by the Mistral7B and GPT4 language models. By utilizing a random forest classifier, we were able to effectively classify relationships into clusters, outperforming mean and median-based clustering methods.

Our approach begins with generating embeddings for relationships between entity pairs using MPNet, which was determined to be the most effective model for capturing textual relationships. We then applied a supervised classification strategy to assign relation labels, leveraging Mistral7B's ability to predict relationships based on context. The approach uses a combination of **Mistral7B/GPT4** for relation extraction, **MPNet embedding model** for relation embeddings and **Random Forest classifier** for classification of relations to clusters.

We validated our methodology on both the private Jio news summary dataset and the public CoNLL04 dataset, demonstrating its adaptability and robustness across different types of data. The classification strategy using the random forest classifier emerged as the best-performing

approach, yielding superior precision, recall, and F1 scores compared to other methods and classifiers.

Overall, our work highlights the potential of combining generative models with traditional classifiers for relation extraction tasks. By leveraging the strengths of both approaches, we were able to create a more nuanced and accurate model that advances the field of relation extraction. This study provides a foundation for future research and applications in natural language processing, particularly in the integration of diverse models to enhance the extraction of complex relationships from text.

8. Future Work Possibilities

Model Tuning: Further tuning of the Random Forest classifier (experiment with hyperparameters or train on larger datasets) and embedding generation process could potentially improve performance.

Extension to Other Datasets: Applying this methodology to other relation extraction datasets to assess the generalizability of the approach.

Incorporation of Advanced Language Models: While this study majorly utilized Mistral7B for generating relation predictions and GPT4 while testing on the public dataset, we could further explore the use of other advanced language models, to understand their impact on relation prediction accuracy.

Fine-tuning LLMs: Instead of prompting LLMs, we can try to fine-tune small LLMs on our specific task, starting with a few samples starting with 500. However, the challenge here would be having sufficient GPU memory to train.

9. References

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